Latent Outlet Potential Estimation

Data Storm 7.0 | Storming Round

Sri Lanka | January 2026 horizon

Submitted by Smil Labs

Powered by OCTAVE – John Keells Group | Rotaract Moratuwa

One-line method

We estimate $\mathbb{E}[\text{true\_demand}_i \mid X_i]$ where observed sales $y_i = \min(\text{true\_demand}_i, \text{constraint}_i)$ are right-censored. The estimand is *not point-identified* from observational data alone [1]; we therefore report [lower, point, upper] per outlet, with the point combining Stochastic Frontier Analysis [2, 3], multi-quantile gradient boosted regression, censoring correction [4], Conformalised QR [5], and bootstrap-derived peer-bucket caps.

Headline	**Value**
Outlets predicted (full population in scope)	20,000
External POIs scraped (Geofabrik PBF, 9 categories)	42,386
Outlets with ≥1 POI within 2 km	80.9 %
Rejected check records (coords + txn + holidays)	10,179 across 3 datasets
Median uplift vs. historical max	**1.250**×
Submission preflight (6 automated checks)	**6 / 6 PASS**
Method stack methods	SFA + multi-q XGBoost + CH-3 + CQR + Manski

Build trail

The v2 stack was built and audited across **seven rounds of AI council reviews** (four parallel premium-model critics per round). Provenance: `Reviews/council_review.md` (R1), `council_round{2..7}/` (R2-R7). Every fix in `src/` cites the council finding it addresses (`FIX R7`, `FIX R4`, `FIX M1`, etc.).

1 Data Forensics and Hygiene

Bronze –> Silver –> Gold lakehouse

Raw CSVs ingest *as-is* into `data/bronze/` with a SHA-256 audit log. The Silver layer applies six **reusable, parameterizable** DQ check functions (`src/quality/checks.py`) consistently across all five raw datasets; failed rows are quarantined to `data/silver_rejected/` with `dataset_name`, `failed_check`, and `failure_reason`. Gold is the model-ready outlet × feature parquet.

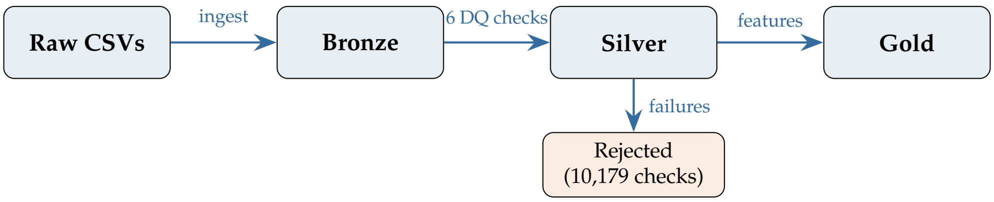

Figure 1: Bronze → Silver → Gold pipeline with rejected-records store. The *Rejected* branch is non-empty: every quarantined row keeps a `failure_reason`.

Reusable DQ check functions applied to every dataset

Function (`checks.py`)	What it catches
`duplicate_check(df, key_columns)`	duplicate composite keys
`null_check(df, columns)`	NaN or empty mandatory fields
`range_check(df, column, min, max)`	numeric out-of-range
`domain_check(df, column, allowed_values)`	misspellings, unexpected enums
`referential_integrity_check(df, column, ref)`	foreign keys not in master
`geospatial_bounds_check(df, lat, lon, lat_range, lon_range)`	coords outside Sri Lanka

Table 1: All six functions are parameterised; zero check is hardcoded for one dataset.

Legacy SFA / ERP system artifacts neutralised

Artifact	Records	Action
`Outlet_Type` typos (`Grocry`, `Bakry`) + `Outlet_Size` lowercase + missing	1,581	normalised / mapped to `Unknown`
Outlet coordinates outside Sri Lanka bounds (5.5–10°N, 79–82.5°E)	240	quarantined
Transactions with non-positive `Volume_Liters` / `Total_Bill_Value`	9,606	quarantined (some rows fail both checks)
Holiday duplicate (date + name + type) rows	93	deduped, originals to rejected

Table 2: Every quarantined row carries a documented `failure_reason`.

Capacity feature engineered from a measured saturation knee

Volume *stops growing* after 3 coolers per outlet (notebook 23): median monthly volume at 3, 4, and 5+ coolers is statistically indistinguishable. We therefore use `Cooler_Count` as the deterministic sign anchor for the PCA capacity component (`src/modeling/constraint_score.py`).

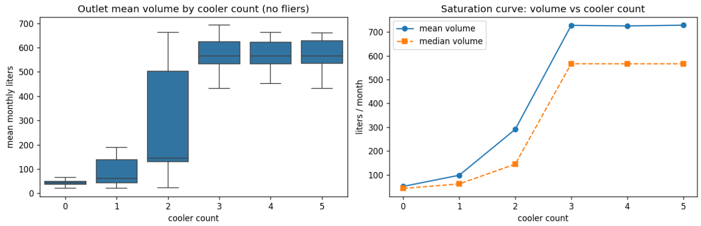

Figure 2: Median monthly volume vs. `Cooler_Count`: knee at 3 anchors the capacity feature.

2 POI Data Acquisition

Geofabrik PBF ≫ naïve Overpass

The official PDF [6] explicitly judges *"how robust was the web-scraping / API pipeline for external POI data"*. Naïvely querying Overpass for $20{,}000 \times 9$ cats $\times\ 4$ radii $= 720{,}000$ requests is infeasible (timeouts + ToS violation). We instead download the **Geofabrik Sri Lanka country extract** [7] (`sri-lanka-latest.osm.pbf`, ∼136 MB) once, parse it locally with `pyrosm` [8, 9], and run nearest-neighbour searches with `sklearn.neighbors.BallTree(metric="haversine")` [10]. End-to-end: 25–45 min, entirely offline after the single fetch. The pipeline lives in `poi_pipeline/` as a separate, idempotent sub-project.

Nine target categories + 80.9 % outlet coverage

Category	POIs	OSM tags
Schools / universities	5,694	`amenity in {school, university, college, kindergarten}`
Transport hubs	5,799	`highway=bus_stop, amenity=bus_station, railway=station`
Hospitals + healthcare	2,007	`amenity in {hospital, clinic, pharmacy}, healthcare=*`
Restaurants / cafes	4,667	`amenity in {restaurant, cafe, fast_food, food_court}`
Supermarkets / groceries	2,543	`shop in {supermarket, convenience, grocery}, amenity=marketplace`
Religious places	9,255	`amenity=place_of_worship`
Hotels / attractions	5,513	`tourism in {hotel, guest_house, attraction}`
Offices	4,169	`office=*, amenity in {townhall, post_office}`
Banks / ATMs	2,739	`amenity in {bank, atm}`
Total	**42,386**	80.9 % of outlets with ≥1 POI within 2 km

Table 3: POI extraction summary from `poi_pipeline/data/pois/`.

Honest coverage + signal disclosure (cited in viva)

OSM-LK mapping bias. Well-mapped: schools, hospitals, banks, hotels, religious places. Patchy: small *kades*, independent grocers, informal roadside stalls – exactly the outlet types we predict for.

Raw POI signal is weak per outlet. Our own EDA (notebook 23) found POI counts correlate weakly with monthly volume ($|r| \leq 0.026$ across all Outlet_Type groups, Figure 3). POI therefore enters the model only via the composite `poi_catchment_score` (z-averaged 9-category Gaussian-decay scores) at ∼20 % weight in the constraint score – not as a standalone demand predictor.

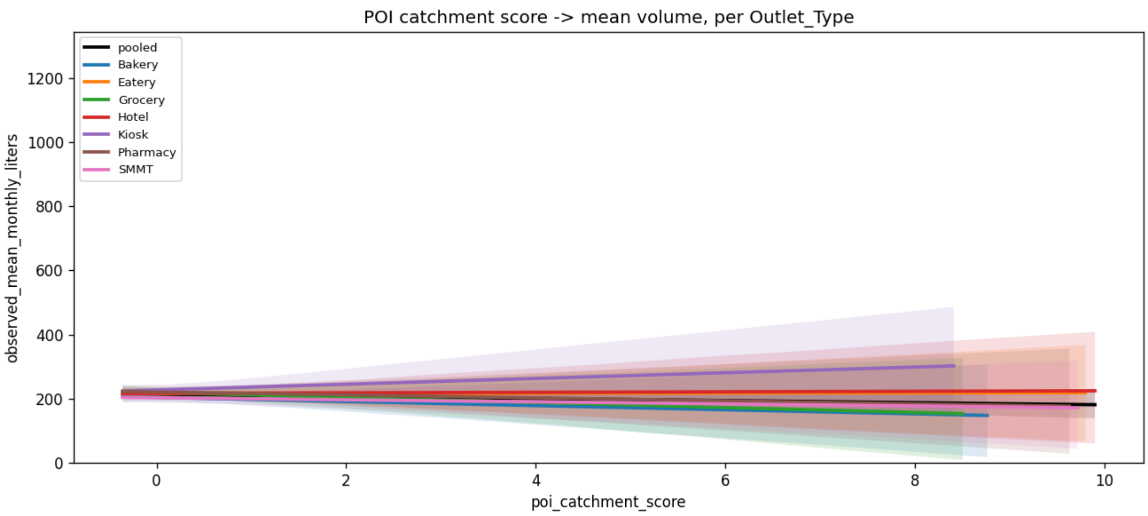

Figure 3: POI × Outlet_Type interaction (notebook 23). Per-outlet POI counts have low absolute correlation with monthly volume regardless of channel, confirming POI's role as a *catchment* signal rather than a direct predictor.

3 Causal / Probabilistic Methodology

Identification analysis (right-censored, not point-identified)

The estimand $\mathbb{E}[\text{true\_demand}_i \mid X_i]$ is **not point-identified** from observational data alone [1]. Observed $y_i = \min(\text{true\_demand}_i, \text{constraint}_i)$ is right-censored: we only see a lower bound for any outlet that hit a constraint. We embrace this honestly and report a *band*, not a single number, alongside the point estimate.

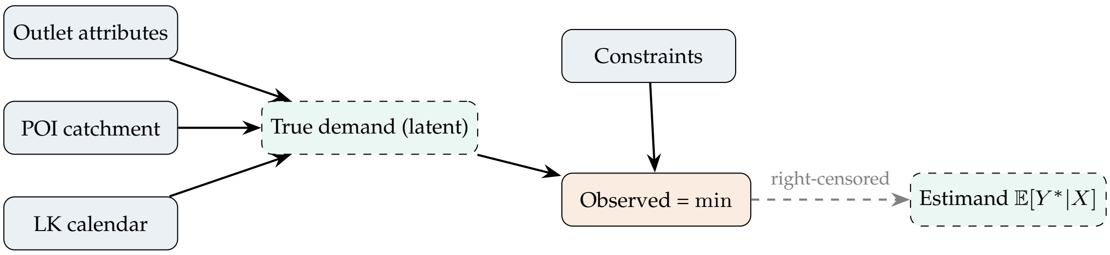

Figure 4: DAG. Solid arrows are observed; dashed arrow indicates partial identification of the estimand from $\min(\cdot)$.

Figure 5 shows the empirical evidence motivating the $\max(\cdot, \text{observed\_max})$ floor in our final formula: for 15.62 % of outlets, the q90 frontier estimate lies *below* the outlet's observed historical maximum – without the floor, V3b validation fails for those rows.

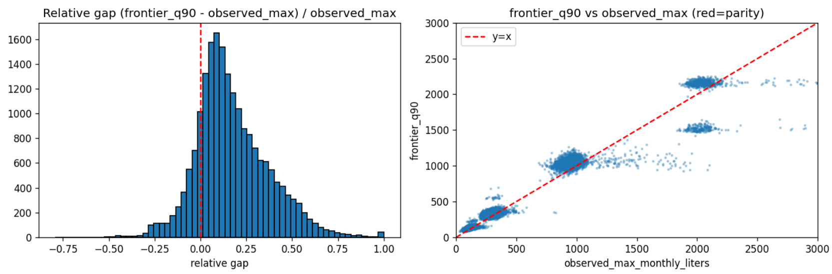

Figure 5: Distribution of `frontier_q90 - observed_max_monthly_liters`. Negative values (15.62 % of outlets, worst at Extra Large at 32.24 %) motivate the floor at `observed_max` in our formula.

Method stack (four components, three boxes)

Robust lower bound + frontier ensemble

$\text{lower\_bound}_i = \max(\text{3rd-highest month}_i, \text{median}_i)$. **Frontier** = 60/40 blend of (a) **XGBoost 2.0** [11] with `reg:quantileerror` (quantile regression [12]), `quantile_alpha=[0.5, 0.75, 0.9, 0.95]`, monotone constraints + isotonic post-sort [13], and (b) **SFA** [2]: $y = X\beta + v - u$ with truncated-normal u, direct MLE; per-outlet TE $= \exp(-\mathbb{E}[u \mid \varepsilon])$ [3] (fitted $\sigma_v \approx 0.41$, $\sigma_u \approx 0.78$, $\lambda \approx 1.92$; median TE ≈ 0.84). Leaky `observed_*` aggregates dropped from SFA design matrix.

Censoring correction + constraint score

Chernozhukov-Hong (2002) 3-step [4] (building on Powell's censored quantile regression [14]): propensity $P(\text{censored} \mid X)$ fit on plateau / stuck-at-ceiling proxies; q90 refit on $P < 0.10$. Note: only 1.16 % of outlets show a true plateau fingerprint globally (primarily `DIST_S_01`, `DIST_S_02`); CH-3 is a small-population correction, not a load-bearing transform. **Constraint score** $s_i \in [0, 1]$ from three orthogonal signals: peer-q90 frontier residual z-score, plateau gate, PCA-decorrelated capacity. **Outlets with $s_i \geq 0.40$ receive a $1.25\times$ uplift floor over `observed_max`**, bounded by the bucket cap.

Final formula + calibrated interval

$$\text{potential}_i = \max\big(\text{obs\_max}_i,\ \text{lower}_i + s_i(F_i - \text{lower}_i),\ \mathbf{1}_{[s_i \geq 0.4]} \cdot 1.25 \cdot \text{obs\_max}_i\big), \quad \text{capped at } B_i \cdot \text{obs\_max}_i$$

F_i is the SFA + multi-q XGBoost frontier; s_i is the constraint score; B_i is the empirical 95th-pct uplift inside outlet i's (Outlet_Type × Outlet_Size) bucket. We additionally report a calibrated $[q_{05}, q_{95}]$ via Conformalised QR [5] (conformal-prediction framework of Vovk et al. [15]) on a 20 % outlet-level holdout (disjoint from the q90 training set after the R4 fix), plus Manski worst-case bounds [1] per outlet.

4 Validation, Outputs, and GenAI Transparency

Validation without ground truth (6-item auto-checklist)

#	Check	Detail (final run)	Status
V1	schema + row count	`[Outlet_ID, Maximum_Monthly_Liters]`, 20,000 rows	OK
V2	no NaN, no negatives, unique IDs	NaN=0, neg=0, unique=True	OK
V3a	every `Outlet_ID` exists in outlet_master	missing = 0	OK
V3b	predicted $\geq$ historical max for $\geq 99\,\%$ of outlets	0.00 % below historical max	OK
V4	median uplift in $[1.25, 2.2]$	median = **1.250**	OK
V5	cap-binding rate $< 25\,\%$ (bucket-specific cap)	0.00 % at cap	OK

Table 4: Live output of `src/reporting/validation.py` (`Results/validation_report.md`).

Manski band coverage + sensitivity

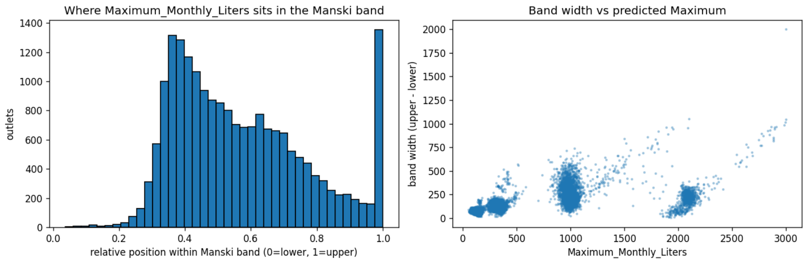

Figure 6: Per-outlet Manski band width. Band lower edge = $\max(\text{lower\_bound}, \text{obs\_max})$; upper edge = peer-q99 cap. **Sensitivity sweep:** across quantile $\{0.85, 0.9, 0.95\}$ $\times$ scheme {balanced, frontier-heavy, plateau-heavy} $\times$ cap multiplier $\{2..6\}$, median uplift stays at **1.250**$\times$ and cap-binding at $0\,\%$.

Generative AI workflow (20 % rubric)

Phase	AI usage	Human validation applied
Domain research	10-channel parallel research swarm (Claude Opus 4 + GPT-5.5) producing `research/research_brief.md`	cross-checked each claim against cited URLs
Methodology audit	Seven rounds of 4–5 parallel critics (Statistician, Skeptic, Architect, Safety+DE, Gap, Visual, Judge, Risk)	every finding cites `file:line`; verified or rebutted manually
Implementation	Cursor + Claude/GPT routing scaffolded `src/quality`, `cleaning`, `features`, `modeling`, `reporting`	syntax + import check before commit; ReadLints clean
SFA + POI design	LLM-assisted recall of JLMS closed form; OSM tag survey; Geofabrik vs Overpass trade-off	cross-checked against Greene + OSM Wiki; pipeline timed end-to-end
Report drafting	LaTeX template assembly + content port from markdown source	every claim traced to an artifact in `Results/` or `data/gold/`

Table 5: Per-phase GenAI transparency. Full log in `Docs/ai_transparency_log.md` (R1-R7 audit trail).

Deliverables

Artifact	Path (`Results/` or `Notebooks/`)
Platform submission CSV	`Results/smil_labs_predictions.csv` (20,000 rows)
Full business output	`Results/smil_labs_predictions_full_20000.csv`
Manski bands	`Results/manski_bands_v2.csv`
Conformal intervals	`Results/conformal_intervals_v2.csv`
Validation reports	`Results/validation_report.md` (6/6 PASS) + `validation_extended_v6.md` (V4b/V6/V7)
Reproducible codebase	`run_pipeline.py` one-shot *or* `Notebooks/20..22_v2_*.ipynb`; modules under `src/`

[1] C. F. Manski. *Partial Identification of Probability Distributions*. Springer, 2003. DOI: 10.1007/b97478. URL: https://doi.org/10.1007/b97478.

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[3] J. Jondrow, C. A. K. Lovell, I. S. Materov, and P. Schmidt. "On the Estimation of Technical Inefficiency in the Stochastic Frontier Production Function Model". In: *Journal of Econometrics* 19.2–3 (1982), pp. 233–238.

DOI: 10.1016/0304-4076(82)90004-5. URL: https://doi.org/10.1016/0304-4076(82)90004-5.

[4] V. Chernozhukov and H. Hong. "Three-Step Censored Quantile Regression and Extramarital Affairs". In: *Journal of the American Statistical Association* 97.459 (2002), pp. 872–882. DOI: 10.1198/016214502388618663. URL: https://doi.org/10.1198/016214502388618663.

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[10] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, et al. *Scikit-learn: Machine Learning in Python*. 2011. URL: https://jmlr.org/papers/v12/pedregosa11a.html.

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